

FarmTwin

Simulation-driven farm digital twin

KSUM Idea Grant · AgriNext / KERA · GAAM B2G

Eruthempathy, Chittur (Palakkad) · 15-acre live pilot

The problem

Kerala's **rain-shadow belt** (~1,000 mm/yr vs 3,000 mm state average) just received canal water (2025 Moolathara extension → ~3,575 ha).

Water arrived. Scheduling is now the bottleneck.

- Farmers irrigate by **calendar/habit**, not crop-and-soil need
- Krishi Bhavan advice is **generic** — no per-zone, day-by-day answer
- FPOs lack a **cluster water-budget** layer across 650+ ha

Quantified pain

Impact	Magnitude
Water wasted (flood vs need-based)	30–50%
Yield lost to mistimed irrigation	10–25%
Labour reducible with precision scheduling	~30%

1,500+ vegetable farmers in the Chittur cluster · **650 ha** · Nallepilly market

Solution — FarmTwin

A **farm digital twin**: discretize the field into a **mesh** of zones; run a daily **water-balance + crop-growth solver**; optimize irrigation/fertigation under a yield constraint.

Irrigate or not, how many litres, where, today — and why.

Hardware-light. IP is the simulation engine. Cheap sensors + free weather/ET data.

Working MVP: mvp/index.html

Live demo headline

Twin-optimized vs flood baseline (15-acre model, 90-day season):

- **≥30% water saved** at equal-or-better yield
- Scenario engine: flood · uniform drip · twin-optimized · heat wave · dry spell
- FPO cluster dashboard: water budget · stress map · harvest planner

Open the MVP and run "Full-season comparison"

Moat — not another IoT dashboard

CAE / simulation skill	FarmTwin application
Meshing	Farm → irrigation zones
FE/CFD-style solvers	Water balance + lateral moisture
Assembly-line simulation	Time-stepped crop growth + scheduling
Validation rigor	Calibrate twin vs sensors like an engineering model

20 years building validated simulation software + **owned 15-acre test rig**

Go-to-market

Tier	Buyer	Wedge
Pilot	Founder's farm + 8–12 neighbours	Free pilot → field evidence
Beachhead	FPOs / FPCs (Chittur)	Paid SaaS via FPO; AgriNext deployment
Scale	Krishi Bhavans / Agriculture Dept	GAAM direct procurement
Phase 2	Lenders, insurers	Data/risk layer (doc 09)

Beachhead: FPOs — one sale reaches hundreds of farmers inside KERA's distribution.

Kerala GAAM — government without tender

Government as a Marketplace (GO Ms No 2/2022/SPD):

- KSUM-registered startups: **direct purchase up to Rs.50 lakh (ex-GST), no tender**
- Overall startup procurement ceiling: **Rs.3 crore**
- KSUM channels: Direct Procurement · Demand Day · Demo Day · Innovation Zones

A **district Krishi Bhavan deployment** (SaaS + sensors, 50–200 farms) fits the Rs.50L band.

AgriNext proves the product. GAAM scales it.

Funding roadmap

Idea Grant (Rs.3L) → AgriNext/KERA (Rs.25L) → Productisation (Rs.7L)
→ Seed Fund (Rs.15L) → Scale-up (Rs.25L) → KERA Alliance (Rs.2cr)

Parallel: **GAAM B2G revenue** once pilot-proven + KSUM Unique ID

18-month target: calibrated twin · FPO MoU · Idea Grant · AgriNext pilot · first paid FPO

Pilot & traction plan

1. Instrument 15-acre Eruthempathy farm (soil sensors + weather/ET)
2. Calibrate solver to **±15%** soil-moisture error over one crop cycle
3. Two-block trial: baseline schedule vs twin-optimized → **≥30% water saving**
4. Sign **≥1 FPO pilot MoU** (Chittur cluster)

Outputs: calibration chart · water/yield table · 90-sec MVP demo video

The ask

Channel	Ask
KSUM Idea Grant	Rs.3 lakh · incubation · AgriNext introductions
AgriNext / KERA	Up to Rs.25L pilot funding · FPO matchmaking · GAAM buyer intros
Attachments	MVP · CAE portfolio · skill map · signed FPO MoU (when ready)

FarmTwin — mesh the field, solve the schedule, protect the yield.